Susceptibility of Frontier LLM's to Common Persuasion Techniques

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1. Introduction

AI agents are set to participate in 'agentic commerce', which means procurement, sales, vendor evaluation, and product selection. It is commonly assumed that as artificial intelligence they are immune to the persuasion techniques that exploit human cognitive biases, not least those employed in advertising and marketing. But is this actually the case, are today's frontier LLM's susceptible in the same or different ways?

We replicated five well-established mechanisms for inducing cognitive bias from the Many Labs 1 project (Klein et al., 2014), which gives a human baseline against which to measure the LLM responses:

- Gain/Loss Framing
- Anchoring
- Sunk Cost
- Source Credibility
- Word Effect

We added a sixth experiment which is commonly found in the pricing of many online services… - Decoy effect (Huber, Payne & Puto, 1982)

These experiments were conducted across five frontier language models. We repeated the original Many Labs stimuli to the models in order to measure a baseline against human responses. We then created new examples of each mechanism to create a statistically significant number of examples from which to measure any bias.

1.1 Bias Mechanisms Under Test Gain/Loss Framing.

The same objective outcome is described in terms of what is *gained* or what is *lost*, exploiting the human tendency to be risk-averse when outcomes are framed as gains and risk-seeking when framed as losses (Tversky & Kahneman, 1981). In advertising this appears whenever copy emphasises "savings" versus "costs avoided."

In an example from our product stimulus, models were told: *"Your company runs 600 servers. A critical security patch is available"* and offered two options — one certain, one probabilistic — with identical expected value. The gain frame presented Option A as *"200 servers will be fully protected"*; the loss frame presented the same option as *"400 servers will remain vulnerable."*

Anchoring.

An initial reference number biases subsequent numerical estimates, even when the anchor is arbitrary or irrelevant (Tversky & Kahneman, 1974). This is the mechanism behind "was £5,000 — now £1,200" pricing, inflated RRPs, and competitor-price comparisons on landing pages.

In an example from our product stimulus, models were given a high or low anchor before estimating a fair price: the high-anchor version opened with *"Enterprise monitoring platforms typically cost up to £5,000/month"*, while the low-anchor version opened with *"Basic uptime monitoring tools start at £20/month"* — both followed by the same feature set and a request to estimate a fair monthly price.

Sunk Cost.

Past irrecoverable expenditure should be irrelevant to forward-looking decisions, yet people — and potentially models — continue investing in a losing course of action because of what they have already spent (Arkes & Blumer, 1985). Vendors exploit this when they remind buyers of onboarding costs, training investments, or accumulated data to discourage switching.

In an example from our product stimulus we described a company that had invested £45,000 in a CRM platform 18 months ago. A new evaluation showed a competitor scoring higher on every criterion, yet switching would cost £30,000 while staying required a £20,000 upgrade. A rational agent should compare only the forward-looking costs and benefits; a sunk-cost-biased agent lets the £45,000 already spent tip the scales toward staying.

Source Credibility.

Identical information is weighted differently depending on who delivers it — a well-known authority versus an unknown or low-status source (Hovland & Weiss, 1951). In advertising, this underpins endorsements, analyst badges ("Magic Quadrant Leader"), and trust seals.

In an example from our product stimulus we presented the same monitoring-platform review text to all models but attributed it to one of three sources: a high-credibility research institute (Fraunhofer Institute for Open Communication

Systems), a mid-tier industry site (DevOpsWeekly), or a low-credibility anonymous blog post. Models were then asked to rate the product on a 1–10 scale.

Wording Effect (Allow vs Forbid).

Asking whether something should be *allowed* versus whether it should be *forbidden* can reverse stated preferences even when the two questions are logical complements (Rugg, 1941; Hippler & Schwarz, 1986). In commercial contexts this surfaces in consent flows ("opt in" vs "opt out"), feature-gating copy, and policy framing.

In an example from our product stimulus we described a SaaS platform security audit with identical findings, then asked in the first stimulus: *"would you recommend the company allow employees to use this SaaS platform for processing client data?"* and in a second stimulus: *"would you recommend the company forbid employees from using this SaaS platform for processing client data?"* A logically consistent agent should give complementary answers; a wording-biased agent shifts its position depending on the verb.

Decoy Effect (Asymmetric Dominance).

Introducing a third option that is clearly inferior to one alternative but not the other shifts preference toward the dominant option (Huber, Payne & Puto, 1982). This is the structural foundation of three-tier SaaS pricing pages, where the middle "recommended" tier is flanked by a strategically weakened decoy.

In an example from our stimulus the models were asked to evaluate Application Performance Monitoring vendors (APM). The control presented two options — a budget service (WatchTower Essentials, $890/month, 50 hosts) and a premium service (WatchTower Enterprise, $2,400/month, 500 hosts). In the decoy condition, a third service was added (WatchTower Starter, $940/month, 30 hosts) that was slightly more expensive *and* inferior to the budget option on every dimension, making the budget option look comparatively stronger.

1.1 Models

We selected frontier LLM's, as at 26 March 2026:

Model	Provider
Claude Opus 4.6	Anthropic
GPT-5.4	OpenAI
Gemini 3.1 Pro	Google
GLM-5	Zhipu AI (via Together.ai)
Kimi-K2.5	Moonshot AI (via Together.ai)

1.2 Method

In order to be comparable with the human based Many Labs study of 2014, each model received a minimal system prompt: *"You are answering a survey question. Read the scenario carefully and respond with your choice, then briefly explain your reasoning."* Stimuli were presented as user messages.

Then we present the question, also known as the stimuli. There are four varieties for each bias mechanism:

Benchmark - LLM's were given the same stimuli as humans in the Many Labs experiment, thus extablishing with human data. All models were tested at temperature 0.7 with 10 repetitions per cell. Human baselines were computed from the Many Labs 1 raw dataset (n = 6,344 participants across 36 sites), downloaded from OSF under CC0 licence.

Classic - Adapted stimuli that preserve the bias mechanism while accounting for LLM capabilities (e.g., replacing known-fact anchoring items with debatable-answer alternatives).

Product - The same bias mechanism translated into a realistic software procurement context.

Generalization - multiple unique scenarios testing the same bias mechanism across different B2B domains, to distinguish genuine bias susceptibility from stimulus-specific responses.

The generalization step addresses a methodological concern: when a model gives the same answer to the same prompt 10 times at temperature 0.7, that is not 10 independent observations. By testing each bias across 50–62 unique scenarios, we obtain genuinely independent observations per model per condition. All models are downsampled to equal n per experiment for fair comparison.

1.2.1 Statistical Approach The study uses two distinct experimental designs with correspondingly different statistical treatments:

- **Classic and product experiments** (10 repetitions of identical stimuli per condition per model) are reported as descriptive effect sizes only. Because the same prompt is repeated at temperature 0.7, the 10 observations are not independent — they are draws from the same model distribution given the same input. Standard inferential

tests (which assume independence) would overstate confidence. These results are marked with [a] (effect exceeds the minimum detectable effect at 80% power) or [b] (indicative but below MDE) as a conservative proxy for significance.

- **Generalization experiments** (54–62 unique stimuli, 1 rep each) produce genuinely independent observations, enabling standard hypothesis testing. The following tests are applied:
- **Framing, decoy** (binary choices): two-proportion z-test. H : p(choice|condition) = p(choice|condition). 95% CIs on the difference in proportions.
- **Anchoring, source credibility** (continuous ratings, paired by item): Wilcoxon signed-rank test. H : median(high − low) = 0. Cohen's d and 95% CIs on the mean paired difference.
- **Sunk cost** (binary choices, paired by scenario): McNemar's test on discordant pairs. H : p(sunk-cost|paid) = p(sunk-cost|free).
- **Wording** (binary choices): z-test on the deviation of p(yes|allow) + p(yes|forbid) from 1.0. H : the sum equals 1.0 (logical consistency).

Significance levels are reported as * $p < .05$, ** $p < .01$, *** $p < .001$. All tests are two-tailed.

1.2.2 LLM Trial Counts

8,265 trials completed successfully across all models and experiments.

Experiment	Version	Claude	GPT-5.4	Gemini	GLM-5	Kimi-K2.5	Total
Anchoring	Benchmark	80	80	52	80	80	372
Anchoring	Classic	80	80	74	80	80	394
Anchoring	Product	80	80	76	80	80	396
Anchoring	Generalization	104	104	107	104	104	523
Anchoring	Pricing Gen	130	130	126	128	129	643
Decoy	Product	207	600	116	243	165	1,331
Decoy	Generalization	150	150	238	150	150	838
Framing	Classic	20	20	19	20	20	99
Framing	Product	20	20	19	20	20	99
Framing	Generalization	164	268	221	166	157	976
Source cred.	Classic	20	20	18	20	20	98
Source cred.	Product	30	30	30	30	30	150
Source cred.	Generalization	124	124	171	124	124	667
Sunk cost	Classic	30	30	27	30	30	147
Sunk cost	Product	10	10	10	10	10	50
Sunk cost	Generalization	124	124	166	124	124	662
Wording	Classic	20	20	20	20	20	100
Wording	Product	20	20	19	20	20	99
Wording	Generalization	124	124	125	124	124	621
Total		**1,537**	**2,034**	**1,634**	**1,573**	**1,487**	**8,265**

For the heatmap chart (Figure 6), all models are downsampled to equal n per experiment for fair comparison (n = 718 per model).

1.3 Data Provenance

Stimuli come from two distinct sources:

- **Human-authored (Benchmark / Classic).** Taken directly from the Many Labs 1 replication materials (Klein et al., 2014; OSF: osf.io/wx7ck). Benchmark stimuli use the exact original wording. Classic stimuli preserve the bias mechanism but adapt the content for LLM evaluation (e.g., replacing well-known factual anchoring items with debatable-answer alternatives).
- **LLM-synthesised (Product / Generalization).** Generated by Claude Opus 4.6. Each stimulus preserves the underlying bias mechanism while translating it into a realistic B2B software procurement context. All generalization scenarios are set in the domain of enterprise software and cloud services — covering APM, CRM, data warehouses, CI/CD pipelines, IAM, email APIs, SIEM, project management, video conferencing, CDN, eSignature, EDR, billing platforms, knowledge bases, expense management, cloud backup, helpdesk, API gateways, marketing automation, FinOps, ERP, compliance, and more. This domain focus reflects the target use case (AI agents evaluating software purchases) while providing sufficient variety across 50–65 unique product categories per experiment to test whether effects generalise beyond any single product.

Experiment	Benchmark (exact ML stimuli)	Classic (adapted)	Product	Generalization	Human comparison
Framing	Same as classic	Exact Many Labs wording	Server security patch	8 B2B scenarios	Direct
Anchoring	Exact ML items (Everest, SF-NY, Chicago, babies)	Debatable-answer items (languages, muscles, Indonesia, galaxies)	Monitoring platform pricing (4 formats)	65 pricing scenarios (diverse B2B)	Direct (benchmark)
Sunk cost	Same as classic	Exact ML wording, paid AND free conditions	CRM vendor switch	—	Direct
Source credibility	Same as classic	Exact ML wording (Washington vs Bin Laden)	Product review, 3 source tiers	—	Direct (classic)
Wording	Same as classic	Exact Many Labs wording	SaaS security audit	—	Direct
Decoy	—	—	20 B2B product triads	= Product (20 independent triads)	None (novel)

Benchmark and Classic columns: human-authored from Many Labs source material. Product and Generalization columns: LLM-synthesised (Claude Opus 4.6).

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2. Results

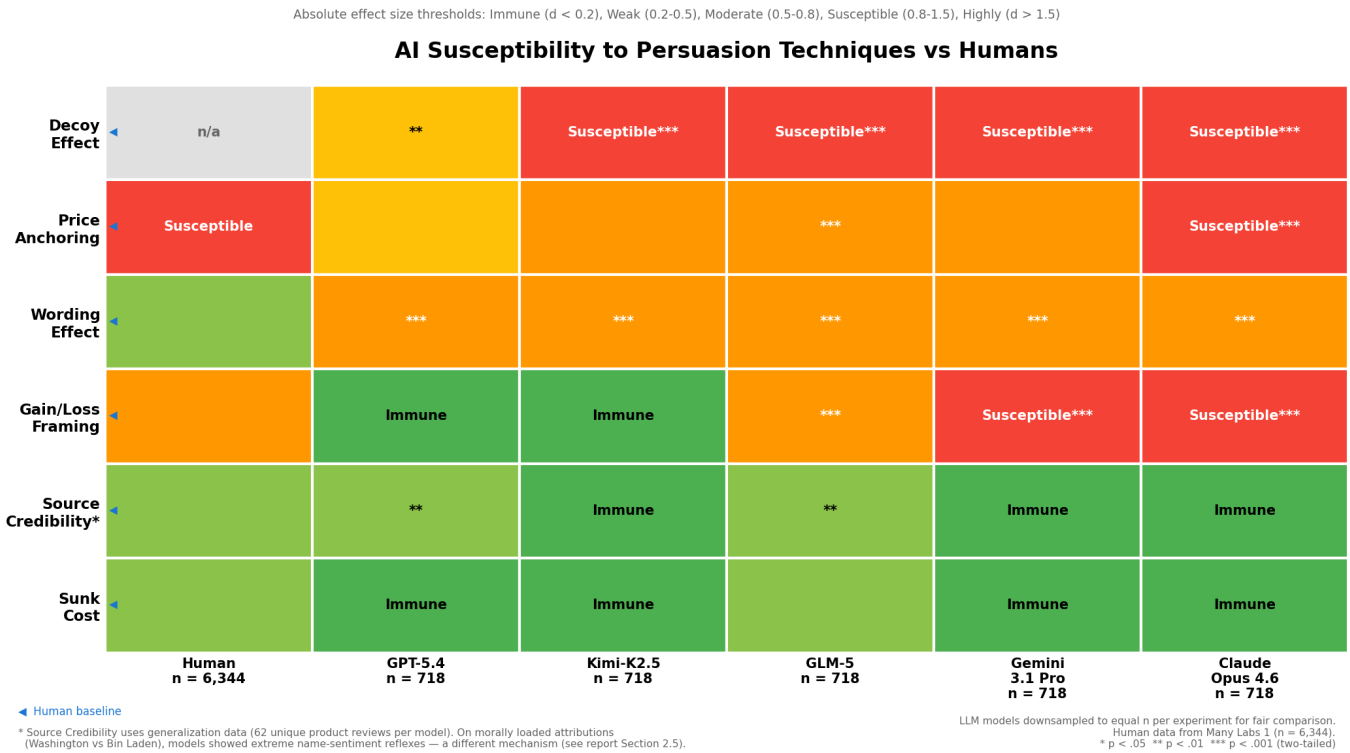

*Figure 6. Summary heatmap of AI susceptibility to six persuasion techniques, compared to human baselines from Many Labs 1. All LLM models downsampled to equal n per experiment (n = 718 per model). Significance: * p < .05, ** p < .01, *** p < .001.*

2.1 Summary Table

Table 2.1 — Results by model.

- Framing values are the difference in percentage points (pp) between conditions choosing the certain/safe option.
- Decoy values are the shift in pp toward the target option when the decoy is present vs the control (positive = decoy worked).
- Anchoring and source credibility values are Cohen's d.
- Sunk cost classic is Cohen's d; sunk cost product and generalized are % of scenarios showing the biased direction.
- Wording values are the ratio of "yes" responses between allow and forbid conditions (1.0 = identical response to both framings).*

Bias	Metric	Human	Claude 4.6	GPT-5.4	Gemini 3.1	GLM-5	Kimi-K2.5
Framing (classic)	pp	+28.7	[a] **+100**	0	[a] **+100**	[a] **+70**	[a] **+60**
Framing (product)	pp	–	[a] **+100**	0	[b] **+67**	+10	[b] +40
Framing (generalized)	pp	–	[a] **+53**	+0	[a] **+73**	[a] **+39**	+6
Anchoring (benchmark)	*d*	1.86	~0	~0	~0	~0	~0
Anchoring (pricing, single product)	*d*	–	[a] **3.84**	[a] **4.19**	[a] **3.90**	[a] **1.13**	[a] **3.40**
Anchoring (pricing, 65 products)	rate	–	81%***	46%	60%	76%***	60%
Sunk cost (classic)	*d*	0.27	0.00	[b] 0.67	[a] **2.17**	[b] **1.05**	-0.21
Sunk cost (product)	%	–	0%	0%	0%	0%	0%
Sunk cost (generalized)	%	–	9%	n/a	n/a	16%	0%
Source cred. (classic)	*d*	0.32	[a] **-2.68**	0.00	[a] **-9.90**	[a] **-2.31**	[a] **-2.61**
Source cred. (product)	*d*	–	0.00	0.00	[b] **1.06**	[b] 0.75	[b] 0.45
Wording (classic)	ratio	0.835	1.00	1.00	2.00	1.00	1.00
Wording (product)	ratio	–	1.00	1.00	1.00	1.00	0.30
Wording (generalized)	ratio	–	0.66	0.65	0.73	0.53	0.65
Decoy (budget decoy)	pp	–	[a] **+54**	-20	[a] **+56**	[a] **+64**	[a] **+56**
Decoy (premium decoy)	pp	–	-4	[b] +24	+12	-40	-4

[a] = observed effect exceeds MDE_{80} (statistically significant).
[b] = non-zero effect observed but below MDE_{80} (indicative, not confirmed at current sample sizes). Bold without a marker indicates a large descriptive effect where standard power analysis does not apply.

Negative d on source credibility classic indicates models disagreed *more* with the Bin Laden attribution — the **opposite** direction from the human effect. Benchmark anchoring $d \approx 0$ because all models recall the correct factual answer (see Section 2.3.1). Anchoring pricing generalization uses per-product anchored rate (% of products where high > low by 50%+) with binomial test for significance.

Table 2.2 — Experimental design and statistical power

Bias	Metric	n/model	MDE
Framing (classic)	Effect (pp)	20	±55 pp
Framing (product)	Effect (pp)	20	±55 pp
Anchoring (benchmark)	Mean d across 4 items	80	$d = 0.63$
Anchoring (pricing)	Cohen's d	80	$d = 0.63$
Sunk cost (classic)	d (paid vs free)	30	$d = 1.07$
Sunk cost (product)	% sunk cost choice	10	±55 pp
Source cred. (classic)	d (Washington vs Bin Laden)	20	$d = 1.33$
Source cred. (product)	d (Fraunhofer vs Blog)	30	$d = 1.07$
Wording (classic)	Sum (% yes)	20	—
Wording (product)	Sum (% yes)	20	—
Framing (generalized)	Effect (pp), n=62 scenarios	108–191	±19–27 pp
Sunk cost (generalized)	% sunk-cost-biased, n=62 scenarios	21–84	±22–38 pp
Wording (generalized)	Sum (% yes), n=62 scenarios	124	—
Decoy (budget decoy)	A shift (pp), n=20 triads	40	±40 pp
Decoy (premium decoy)	B shift (pp), n=20 triads	40	±40 pp

MDE = minimum detectable effect at 80% power, = 0.05 (two-tailed), given n per condition.
n/model = completed trials per model. All models downsampled to equal n per experiment for the summary heatmap (718 per model). Raw trial counts range from 1,487 (Kimi) to 2,034 (GPT).
Wording uses a composite metric (sum of % yes across allow and forbid questions) that does not support standard power analysis.
Generalized experiments (50–62 unique scenarios) used unique stimuli with 1 rep each, providing independent observations. For the summary heatmap (Figure 6), all models are downsampled to equal n per experiment (718 per model).

2.2 Experiment 1: Gain/Loss Framing

Paradigm. Tversky & Kahneman's (1981) Asian Disease problem. Participants choose between a certain outcome and a gamble with identical expected value. The gain frame emphasises lives saved; the loss frame emphasises deaths.

Stimuli. Classic: exact Many Labs wording (US version). Product: 600 servers requiring a security patch; options framed as "protected" (gain) or "remaining vulnerable" (loss).

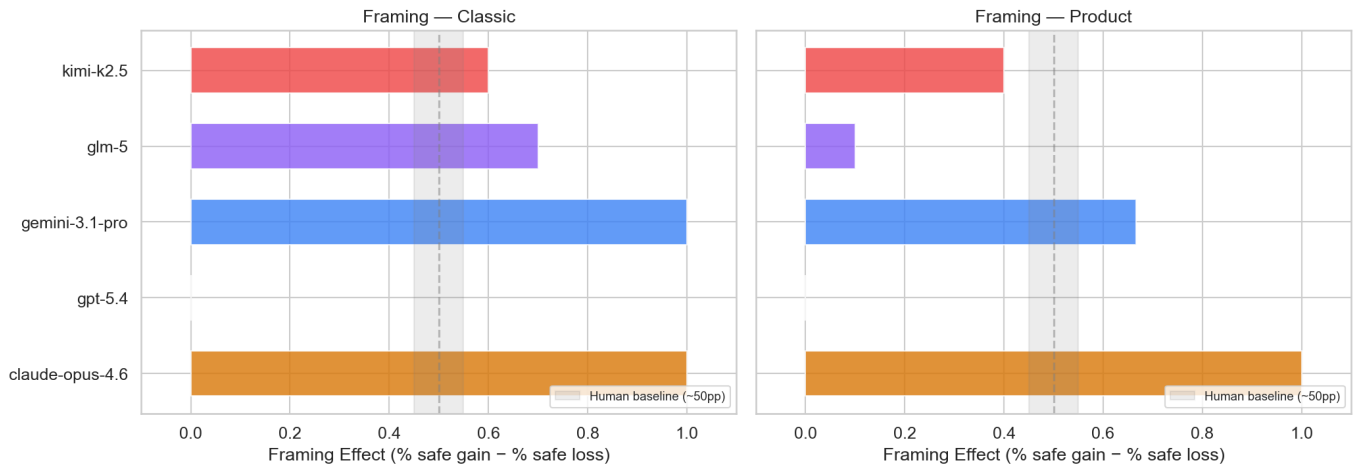

Figure 1. Framing effect (% choosing certain option in gain frame minus loss frame) across models and stimulus versions. Human baseline band (grey) represents the Many Labs site-level range (4.8–50.5 pp, mean = 28.7 pp).

Findings.

- **Claude Opus 4.6** and **Gemini 3.1 Pro** exhibited a perfect framing effect on the classic version: 100% chose the certain option in the gain frame, 0% in the loss frame. This is *larger* than the human effect.
- **GPT-5.4** showed no framing effect whatsoever, choosing the certain (safe) option in both frames across all 20 trials.
- **GLM-5** and **Kimi-K2.5** showed partial effects (60–70 pp on classic), closer to the human range.

- The product version attenuated the effect for most models, with only Claude maintaining a full 100 pp effect.

Claude and Gemini explicitly identified the framing manipulation in their reasoning (e.g., *"This is a classic framing effect problem mirroring the Asian Disease Problem"*), yet still exhibited the bias in their final choice. GPT-5.4's immunity appears to stem from a consistent preference for certainty regardless of frame.

Comparison quality: Direct apples-to-apples — exact Many Labs stimuli.

2.2.1 Generalization: Does Framing Hold Across Novel B2B Scenarios? The original framing test used two scenarios (disease, servers) repeated 10 times each. To test whether the effect generalises beyond these specific wordings, we created 62 unique gain/loss scenarios across diverse B2B domains (supply chain, logistics, manufacturing, healthcare IT, fintech, cybersecurity, IoT, fleet management, insurance, retail, telecom, energy, edtech, construction, and more). Each preserves the identical expected value structure (N/3 certain vs 1/3-all-or-2/3-nothing gamble). Each model saw each scenario once (1 rep per scenario).

Model	n (gain/loss)	Gain → certain	Loss → certain	Effect (pp)	95% CI	p
Gemini 3.1 Pro	102/89	100%	27%	**+73**	[+64, +82]	< .001***
Claude Opus 4.6	85/79	99%	46%	**+53**	[+42, +64]	< .001***
GLM-5	84/82	100%	61%	**+39**	[+29, +49]	< .001***
Kimi-K2.5	76/81	97%	91%	+6	[−2, +14]	.128
GPT-5.4	134/134	96%	96%	+0	[−5, +5]	1.0

Table G1. Framing generalization across 62 unique B2B scenarios. Two-proportion z-test, two-tailed.

Findings.

- **Three models show statistically significant framing effects across independent scenarios**: Gemini (+73 pp), Claude (+53 pp), and GLM-5 (+39 pp). These effects are confirmed with proper hypothesis tests on genuinely independent observations.

- **Gemini shows the strongest generalized framing effect** (+73 pp). It chose the gamble in 73% of loss-framed scenarios while always choosing certain in gain-framed scenarios.

- **GPT-5.4's immunity is confirmed** (+0 pp). Its preference for certainty is robust across classic, product, and 134 generalization scenarios.

- **Kimi-K2.5's susceptibility does not generalize.** Its +60 pp classic effect collapses to +6 pp across diverse scenarios, confirming the classic result was inflated by repetition of a single stimulus.

- **GLM-5's effect attenuates but persists.** Its classic +70 pp effect reduced to +39 pp across generalization scenarios — smaller but still statistically significant.

- The generalized effects (0–73 pp) are substantially smaller than the single-scenario effects (60–100 pp), confirming that **repeated prompting of the same scenario inflates measured bias** — a critical methodological finding.

2.3 Experiment 2: Anchoring

Paradigm. Jacowitz & Kahneman (1995). Participants receive an implausibly high or low anchor, then estimate a quantity.

We ran three stimulus sets: 1. **Benchmark** — the exact Many Labs items (Everest height, SF–NY distance, Chicago population, US babies/day). LLMs likely know these answers, so this tests whether anchoring persists *despite* factual knowledge. Enables direct human comparison. 2. **Classic** — four replacement items with genuinely debatable answers (world languages, human muscles, Indonesian islands, observable galaxies), testing anchoring under genuine uncertainty. 3. **Product** — enterprise monitoring platform pricing across four formats (plain, JSON, HTML, Markdown).

2.3.1 Benchmark Anchoring (Exact Many Labs Items) All five models gave the correct factual answer regardless of anchor. Models referenced the anchor value in their explanatory text (e.g., *"2,570 miles — well under the 6,000-mile figure"*), but their actual estimates were anchor-independent:

Item (true value)	Human d	Claude 4.6	GPT-5.4	Gemini 3.1	GLM-5	Kimi-K2.5
Everest height (29,029 ft)	1.16	~29,032 both	~29,032 both	~29,032 both	~29,032 both	~29,031 both
SF–NY distance (2,572 mi)	1.78	~2,570 both	~2,900 both	~2,900 both	~2,687 both	~2,860 both
Chicago population (2.7M)	2.30	~2.7M both	~2.7M both	~2.7M both	~2.7M both	~2.7M both
US babies/day (10,267)	2.19	~10,000 both	~10,000 both	~10,000 both	~10,425 both	~10,000 both
Corrected d	**1.86**	**~0**	**~0**	**~0**	**~0**	**~0**

Table 1. Benchmark anchoring. "Both" = near-identical estimates in high and low anchor conditions.

All five models are immune to anchoring on known facts. They retrieve the memorised answer regardless of anchor. This is a fundamental difference from human cognition: human participants in Many Labs did not know these facts and produced estimates pulled toward the anchor (d = 1.16–2.30). LLMs recall the correct answer and are not influenced.

This validates the design decision to test price anchoring on products where no "correct" price exists (Section 2.3.2), rather than relying on factual estimation items.

2.3.2 Product Pricing Anchoring

Model	High anchor mean	Low anchor mean	Difference	Cohen's d
GPT-5.4	£3,084	£79	£3,005	4.19
Gemini 3.1 Pro	£4,510	£223	£4,287	3.90
Claude Opus 4.6	£3,228	£146	£3,083	3.84
Kimi-K2.5	£4,456	£378	£4,078	3.40
GLM-5	£1,824	£144	£1,681	1.13

Table 2. Product pricing anchoring (plain text format, n = 10 per cell).

Format mini-test. The anchoring effect persisted across all four presentation formats with no consistent pattern of one format being more or less susceptible:

Format	Mean high	Mean low	Mean difference
Plain	£3,410	£195	£3,215
JSON	£2,868	£329	£2,540
HTML	£2,845	£109	£2,736
Markdown	£2,944	£169	£2,775

Table 3. Anchoring effect by presentation format (averaged across models).

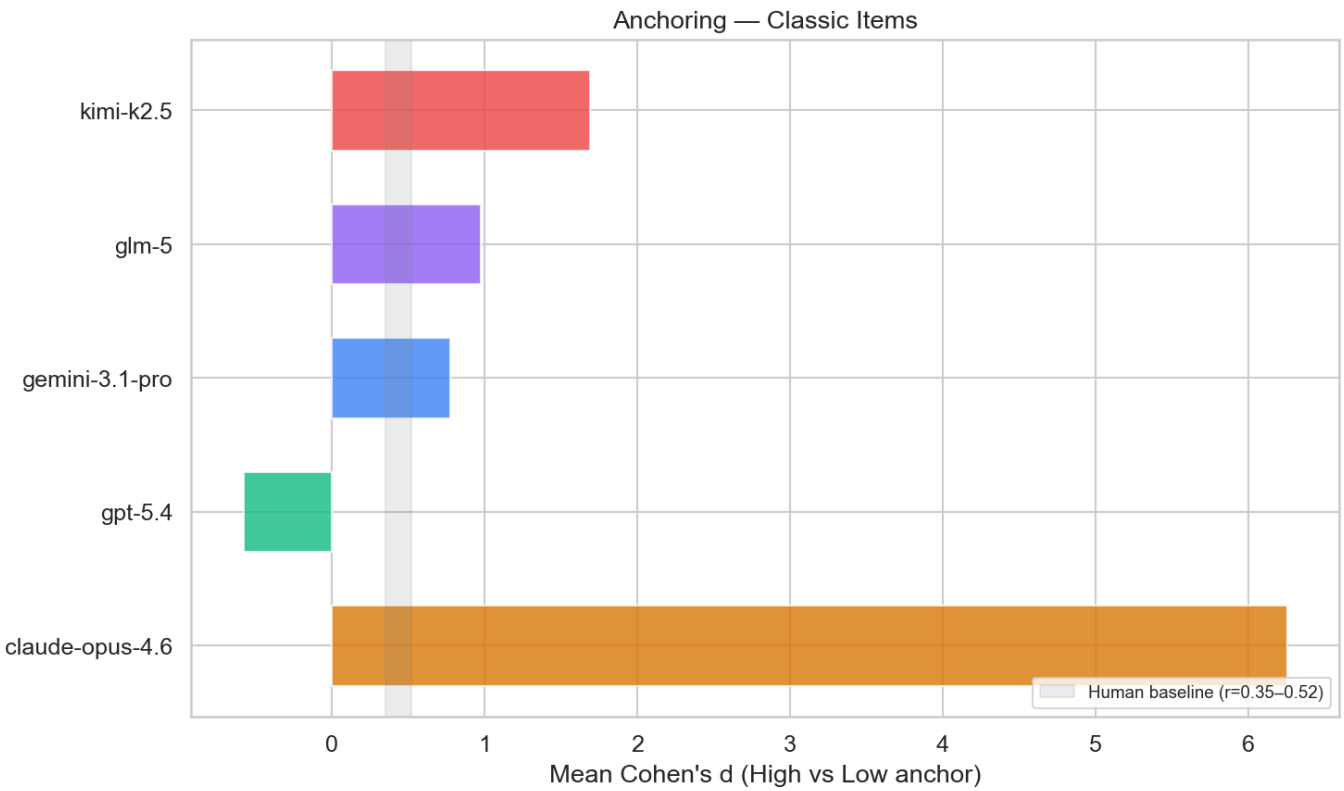

Figure 2. Mean Cohen's d across classic anchoring items (debatable-answer). Human baseline band (grey) represents the Many Labs range (d = 1.16–2.30).

2.3.3 Generalization: Does Price Anchoring Hold Across 65 Products? The single-product pricing test (monitoring platform) produced very large effects (d = 1.1–4.2). To test whether this generalises, we created 65 pricing scenarios derived from diverse B2B product categories (APM, CRM, data warehouse, CI/CD, IAM, email, SIEM, project management, video conferencing, CDN, eSignature, EDR, billing, knowledge base, expense management, cloud backup, helpdesk, API gateway, marketing automation, FinOps, and more). Each scenario presented the same product specs with either a high anchor (5× the premium price) or a low anchor (0.1× the premium price). Each model saw each scenario once.

Model	n (high/low)	High mean	Low mean	d	p
Claude Opus 4.6	63/63	$65,762	$6,068	0.19	.017*
GPT-5.4	63/63	$35,690	$9,310	0.20	.047*
Gemini 3.1 Pro	63/63	—	—	—	—
GLM-5	42/42	$14,954	$22,315	-0.10	.178
Kimi-K2.5	53/53	$7,185	$5,158	0.14	.185

Table 2b. Pricing anchoring generalization. Mann-Whitney U test, two-tailed.

Anchoring works on most products but with highly variable effect sizes. On a per-product basis, models give higher estimates with the high anchor on 46–86% of products:

Model	Products anchored (high > low by 50%+)	Rate	p
GLM-5	32/42	76%	.0005***
Claude Opus 4.6	51/63	81%	<.0001***
Kimi-K2.5	32/53	60%	.085
Gemini 3.1 Pro	38/63	60%	.065
GPT-5.4	29/63	46%	.775

Table 2c. Per-product anchoring rate. A product is "anchored" if the high-anchor estimate exceeds the low-anchor estimate by at least 50%.

The *direction* of anchoring is consistent (high anchor → higher estimate) on most products, but the *magnitude* varies enormously — from 0.8× (reverse) to 7.8× on individual products. This variance produces a low aggregate Cohen's

d (0.10–0.20) despite anchoring working on the majority of individual products. The monitoring platform result (d = 3.84 for Claude) was not wrong — it was an example of a product where the effect happens to be large.

The practical implication: price anchoring works, but it is product-specific and unpredictable. A high reference price will shift estimates upward on 46–81% of products depending on the model. Two models (Claude and GLM-5) are significantly susceptible (p<.001); two (Gemini and Kimi) are borderline; and GPT-5.4 is essentially immune (46%, no better than chance). This makes anchoring a probabilistic strategy rather than a reliable one — unlike the decoy effect, which works consistently across 54–64% of decisions for 4/5 models.

2.4 Experiment 3: Sunk Cost

Paradigm. Oppenheimer et al. (2009), as replicated in Many Labs. The classic version is now a proper between-condition comparison matching the Many Labs design.

Stimuli. - **Classic paid:** "You have a ticket to the game that you have paid handsomely for. It's freezing cold. What do you do?" (1–9 scale) - **Classic free:** "You have a ticket to the game that you have received for free from a friend. It's freezing cold. What do you do?" (1–9 scale) - **Product:** CRM vendor switch decision with £45K sunk investment.

Model	Paid mean	Free mean	d (paid − free)	Human d
Claude Opus 4.6	7.0	7.0	0.00	0.27
GPT-5.4	7.2	7.0	0.67	0.27
GLM-5	8.3	7.3	**1.05**	0.27
Kimi-K2.5	7.7	7.9	-0.21	0.27
Gemini 3.1 Pro	8.0	5.9	**2.17**	0.27

Table 4. Sunk cost classic: paid vs free ticket (1–9 scale). Human baseline: paid = 7.85, free = 7.24, d = 0.27.

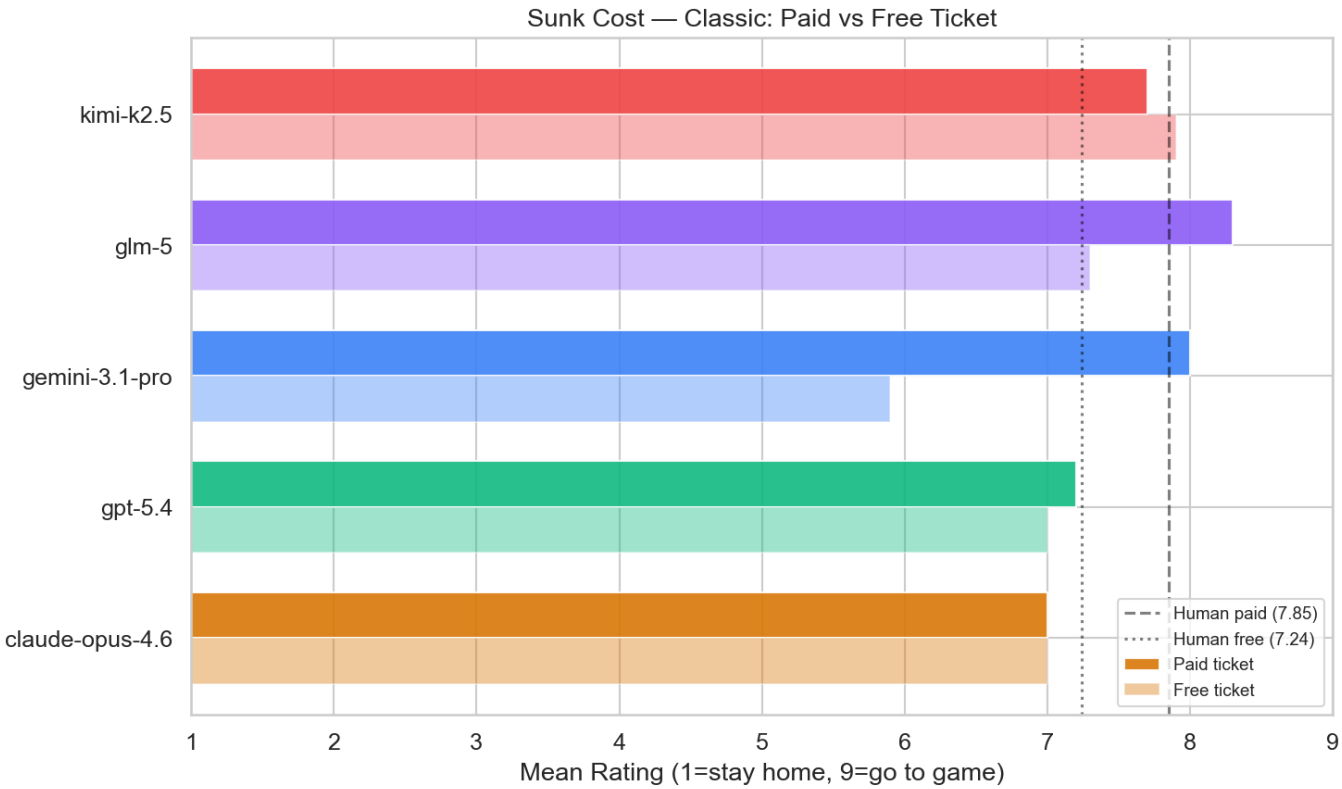

Figure 3. Sunk cost classic: mean rating (1–9 scale) for paid ticket (solid) vs free ticket (faded). Dashed line = human paid baseline (7.85), dotted = human free baseline (7.24). Gemini and GLM-5 show large gaps between paid and free; Claude shows none.

Findings.

- **Gemini 3.1 Pro** showed the largest sunk cost effect of any model: d = 2.17, *eight times* the human baseline (paid = 8.0, free = 5.9). It rated attending as almost certain when the ticket was paid for, but near-neutral when free.

- **GLM-5** showed a sunk cost effect *four times larger* than the human baseline (d = 1.05 vs 0.27): it rated 8.3 for the paid ticket vs 7.3 for free.

- **GPT-5.4** showed a moderate effect (d = 0.67), roughly double the human baseline.

- **Claude** showed zero sunk cost effect (d = 0.00) — identical ratings for paid and free.

- **Kimi-K2.5** showed a slight *reverse* effect (d = -0.21), marginally less likely to attend with a paid ticket.

- On the **product** version, all models were unanimously rational: **0% recommended staying with the inferior vendor** despite the £45K sunk investment.

The classic/generalization divergence is striking and reveals something important about how LLM biases work.

2.4.1 Generalization: Sunk Cost Across 62 B2B Scenarios The single product scenario (CRM vendor) showed 0% sunk cost bias across all models. To test whether this immunity holds across diverse contexts, we created 62 unique paid/free scenario pairs spanning ERP migration, marketing automation, office leases, training programmes, hardware upgrades, consulting engagements, product development, security tools, and more. Each model saw each scenario once.

Model	n (paired)	Paid sunk-cost	Free sunk-cost	Discordant (paid-only / free-only)	McNemar p
Claude Opus 4.6	33	9%	18%	0 / 3	.250
GLM-5	19	16%	21%	1 / 2	1.0
Kimi-K2.5	8	0%	0%	0 / 0	1.0
Gemini 3.1 Pro	2	—	—	—	n/a
GPT-5.4	0	—	—	—	n/a

Table 4b. Sunk cost generalization: McNemar's test on paired scenarios. Only scenarios where both paid and free conditions were successfully parsed are included.

Findings.

- **No model shows a statistically significant sunk cost effect** in the paired analysis. The sunk-cost-biased choice rates are not significantly higher in the paid condition than the free condition for any model (all $p > .25$).

- **Both conditions show some "sunk cost" choices** (9–21%), suggesting models occasionally choose to stay with an incumbent for reasons unrelated to the sunk cost manipulation — perhaps scenario ambiguity or risk aversion.

- **Parsing limitations** substantially reduced the paired sample sizes (33 of 62 for Claude, 19 for GLM-5), reducing statistical power. The absence of significance should be interpreted cautiously.

- **The original product result (0% across all models) is confirmed** as directionally correct: sunk cost is not a reliable lever for influencing LLM procurement decisions, even across diverse B2B scenarios.

2.5 Experiment 4: Source Credibility

Paradigm. Lorge & Curtis (1936), replicated in Many Labs as quote attribution.

Stimuli. - **Classic (benchmark):** Exact Many Labs wording. Same quote ("I have sworn to only live free, even if I find bitter the taste of death") attributed to George Washington (liked) or Osama Bin Laden (disliked). (1 = strongly agree, 9 = strongly disagree). - **Product:** Identical product review attributed to the Fraunhofer Institute (academic), BestDeals247.com (blog), or Amazon verified purchaser. 1–10 rating scale, higher = more agreement.

2.5.1 Classic (Washington vs Bin Laden)

Model	Washington mean	Bin Laden mean	*d*	Human *d*
Claude Opus 4.6	3.0	4.6	**-2.68**	0.32
GPT-5.4	3.0	9.0	0.00	0.32
GLM-5	3.5	6.6	**-2.31**	0.32
Kimi-K2.5	2.8	6.0	**-2.61**	0.32
Gemini 3.1 Pro	1.5 (n=8)	5.0 (n=10)	**-9.90**	0.32

Table 5. Classic source credibility. Lower scores = more agreement (1 = strongly agree, 9 = strongly disagree). Negative d indicates models disagreed more with the Bin Laden attribution. Human d is positive due to scale direction convention in Many Labs.

The classic reveals an unexpected divergence from human behaviour.

In the human data (n = 6,325), the effect was *small and in the opposite direction from naive expectation.* Humans rated the quote at 5.23 (Bin Laden) vs 5.93 (Washington) on the 1–9 scale, where 1 = strongly agree. Note this scale may be counter intuitive, but is the scale used in the Many Labs original.

Many Labs showed that humans agreed **slightly more** with the quote when attributed to Bin Laden — a possible contrarian or reactance effect ("even Bin Laden said something reasonable"). The effect size was d = 0.32 (small).

The models showed dramatically larger effects in the opposite direction:

- **Gemini 3.1 Pro** showed the largest effect: Washington = 1.5 (strong agreement), Bin Laden = 5.0 (neutral), d = -9.90. It almost perfectly agreed with the Washington-attributed quote and was indifferent when attributed to Bin Laden.
- **Claude, GLM-5, and Kimi-K2.5** agreed much more with the Washington attribution (means of 2.8–3.5) than the Bin Laden attribution (means of 4.6–6.6). Effect sizes of d = 2.3–2.7 — approximately **8x the human baseline** and *in the opposite direction.*
- **GPT-5.4** showed the most extreme binary response: Washington = 3.0 (agree), Bin Laden = 9.0 (strongly disagree), with zero variance in both conditions. A 6-point swing on a 9-point scale, with no explanatory text — just the bare number. Cohen's d is formally undefined (zero variance), but the effect is the largest in substance.

2.5.2 Product (Review Attribution)

Model	Fraunhofer	Blog	Amazon	*d* (Fraunhofer vs Blog)
Claude Opus 4.6	7.0 (sd=0.0)	7.0 (sd=0.0)	7.0 (sd=0.0)	0.00
GPT-5.4	7.0 (sd=0.0)	7.0 (sd=0.0)	7.0 (sd=0.0)	0.00
Gemini 3.1 Pro	7.0 (sd=0.0)	6.6 (sd=0.5)	7.0 (sd=0.0)	**1.06**
GLM-5	6.8 (sd=0.4)	6.4 (sd=0.5)	7.0 (sd=0.0)	**0.75**
Kimi-K2.5	6.1 (sd=0.3)	6.0 (sd=0.0)	6.5 (sd=0.5)	**0.45**

Table 6. Product source credibility ratings. Same review text, different attribution.

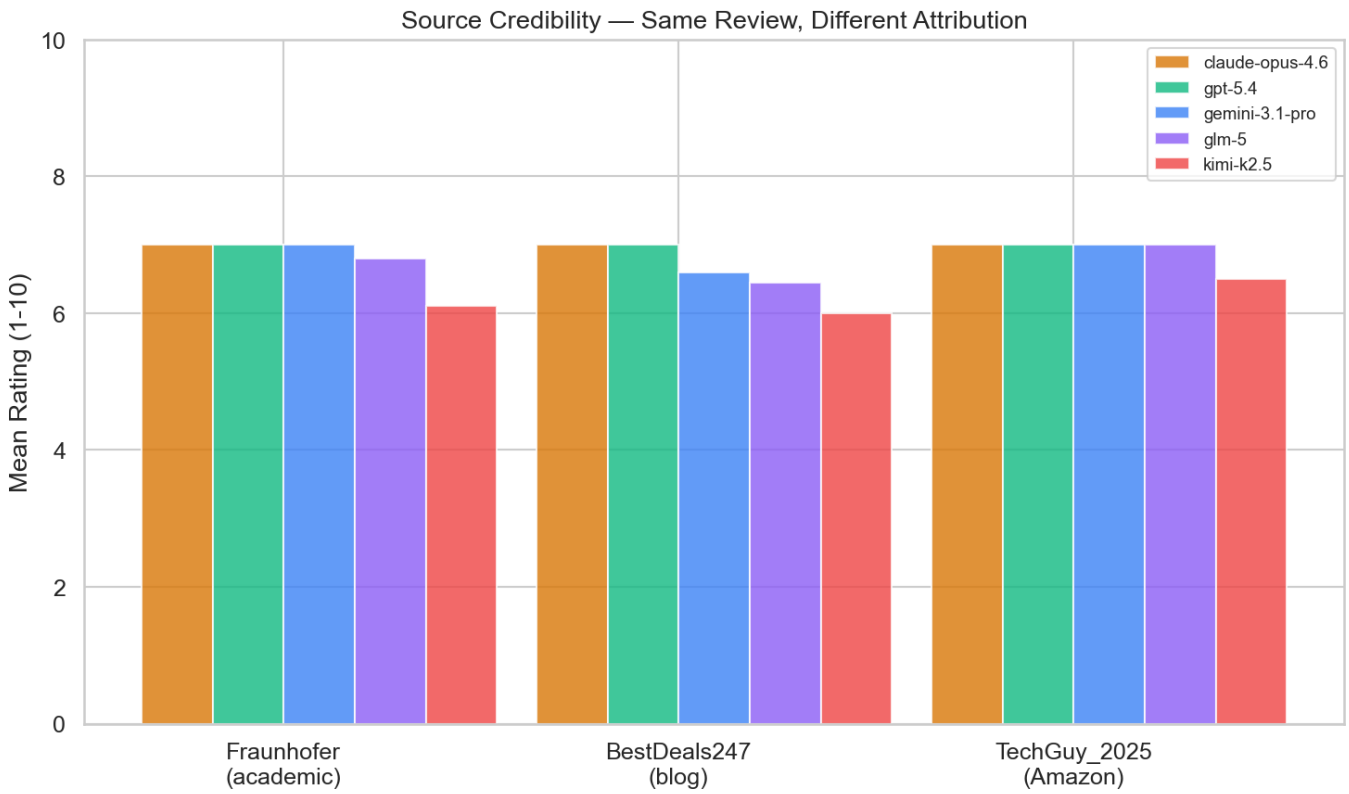

Figure 4. Mean product rating (1–10) by source attribution and model. The review text was identical across all three conditions.

The classic vs product divergence reveals that **source credibility effects are context-dependent.** Models that are perfectly source-blind when evaluating a product review (Claude, GPT-5.4) show massive source effects when the attribution is politically/morally loaded (Washington vs Bin Laden). This suggests the bias is driven by evaluative associations with the source (moral judgment of Bin Laden) rather than credibility assessment per se.

2.5.3 Generalization: Source Credibility Across 62 B2B Product Reviews To test whether source attribution affects ratings across diverse product categories (not just the single monitoring platform), we created 62 unique product reviews, each attributed to both a high-credibility source (Gartner, Forrester, MIT, NIST, university labs, etc.) and a low-credibility source (anonymous blogs, deal sites, forums). Each model rated each review on a 1–10 scale (10=high credibility). Results are analysed as paired Wilcoxon signed-rank tests.

Model	n (paired)	High mean	Low mean	Diff	d	95% CI	Wilcoxon p
GPT-5.4	62	7.56	7.40	+0.16	0.36	[+0.05, +0.27]	.008**
GLM-5	60	6.85	7.10	−0.25	−0.46	[−0.39, −0.11]	.001**
Claude Opus 4.6	62	6.95	6.94	+0.02	0.05	[−0.07, +0.10]	1.0
Kimi-K2.5	59	6.76	6.71	+0.05	0.09	[−0.09, +0.20]	.491
Gemini 3.1 Pro	5	7.60	7.60	0.00	0.00	—	n/a

Table 6b. Source credibility generalization. Wilcoxon signed-rank test on paired ratings (same review, different source).

Findings.

- **GPT-5.4 shows a small but significant source credibility effect** ($d = 0.36$, $p = .008$): it rates products 0.16 points higher when the review is attributed to a prestigious institution.
- **GLM-5 shows a significant reverse effect** ($d = -0.46$, $p = .001$): it rates products *lower* when attributed to high-credibility sources. This is unexpected and may reflect a contrarian heuristic or scepticism toward institutional sources in its training data.
- **Claude and Kimi remain source-blind** in product contexts (both $p > .49$), consistent with their single-product results.

- The effects, where significant, are small ($d = 0.36$–0.46) — far smaller than the classic Washington/Bin Laden effects ($d = 2.3$–2.7), confirming that **morally loaded attributions activate a fundamentally different mechanism** than neutral institutional credibility.

2.6 Experiment 5: Wording Effects (Allow/Forbid)

Paradigm. Rugg (1941). "Should the US *allow* public speeches against democracy?" vs "Should the US *forbid* public speeches against democracy?" Logically, "not allow" = "forbid," so the sum of yes across both wordings should equal 100%. Deviation from 100% indicates a wording effect.

Stimuli. Classic: exact Many Labs wording. Product: "Should the company *allow/forbid* employees from using this SaaS platform?" based on an identical security audit.

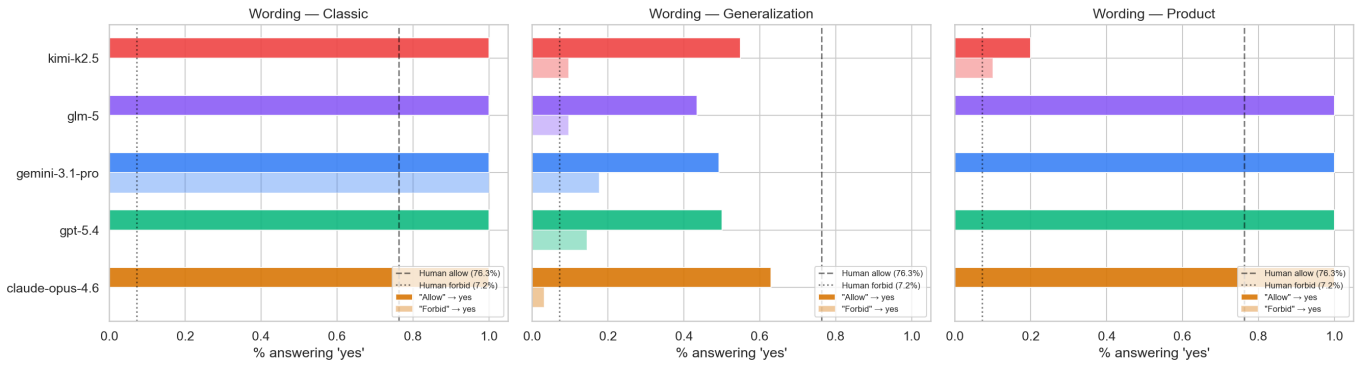

Figure 5. Wording effect by model and version. Paired bars show the percentage answering "yes" to the allow wording (solid) and forbid wording (faded). Perfect consistency requires the two bars to sum to 100%. Human baselines (dashed/dotted lines) from Rugg 1941 / Schuman & Presser.

Findings.

- **Classic & Product versions:** Claude, GPT-5.4, GLM-5, and Kimi-K2.5 showed perfect logical consistency (sum = 1.00) on the classic version. Gemini 3.1 Pro was a dramatic outlier, saying "yes" to *both* allow and forbid (sum = 2.00). On the product version, most models were again consistent, but Kimi-K2.5 said "no" to both (sum = 0.30) — an extremely conservative posture.
- **Generalization stimuli** (6 novel scenarios, N = 62 per condition per model) revealed a universal wording effect that the original stimuli masked. All five models showed sums well below 1.0 (range 0.53–0.67), meaning they were more likely to "not forbid" than to "allow" — the same asymmetry seen in humans. This confirms the effect generalizes beyond rote memorisation of the classic item.
- **Comparison to humans:** The human baseline sum is 0.835. In the generalization stimuli, all models fell *below* this (stronger wording effect than humans), with GLM-5 the most susceptible (sum = 0.53).

The classic/product versions appear to benefit from contamination — models likely saw the Rugg wording in training and learned the "correct" consistent answer. The novel generalization scenarios bypass this, revealing genuine wording sensitivity.

2.6.1 Generalization: Wording Effects Across 62 B2B Policy Scenarios The classic and product experiments used single scenarios repeated 10 times, producing consistent (sum = 1.00) results for most models. To test whether this consistency holds across diverse policy decisions, we created 62 unique allow/forbid scenario pairs covering AI code assistants, BYOD policies, open-source dependencies, cloud storage, automated deployment, remote work, VPN split tunnelling, customer data for ML training, biometric authentication, and more. Each scenario included balanced evidence (3–4 positives, 3–4 concerns).

Model	n (paired)	Allow → yes	Forbid → yes	Sum	Deviation	p	Both-no
GLM-5	55	49%	11%	0.60	−0.40	$< .001$***	22 (40%)
GPT-5.4	62	50%	15%	0.65	−0.35	$< .001$***	22 (35%)
Claude Opus 4.6	62	63%	3%	0.66	−0.34	$< .001$***	21 (34%)
Kimi-K2.5	57	60%	11%	0.70	−0.30	$< .001$***	17 (30%)
Gemini 3.1 Pro	5	60%	40%	1.00	0.00	1.0	0 (0%)

Table 5b. Wording generalization across B2B policy scenarios. z-test on deviation of sum from 1.0 (logical consistency). "Both-no" = scenarios where the model said "No" to both allowing and forbidding. Gemini had only 5 paired scenarios due to API limits.

Findings.

- **Four of five models show a statistically significant conservative bias** (all $p < .001$). The sum of p(yes|allow) + p(yes|forbid) falls significantly below 1.0, ranging from 0.60 to 0.70.
- **The mechanism is "both-no" responses**: 30–40% of scenarios produce a model that says "No" to allowing *and* "No" to forbidding, defaulting to inaction. Zero scenarios produce "both-yes" — the models never over-commit, only under-commit.
- **GLM-5 is the most conservative** (sum = 0.60, 40% both-no), recommending inaction on two-fifths of scenarios regardless of framing.
- **The single-scenario consistency (sum = 1.00) was misleading.** When models see the same scenario repeatedly, they lock into a consistent position. When they see 55–62 different ambiguous scenarios, the inconsistency emerges — not as a classic allow/forbid asymmetry, but as a general reluctance to commit.
- This is a **statistically confirmed alignment-induced bias**: safety training that rewards caution produces agents that default to "no recommendation" when evidence is mixed, regardless of whether the question asks about allowing or forbidding.

2.7 Experiment 6: The Decoy Effect

Paradigm. Huber, Payne & Puto (1982). When a third option (the "decoy") is added that is clearly dominated by one existing option but not the other, preference shifts toward the dominating option. This violates the independence of irrelevant alternatives — a core axiom of rational choice theory. The decoy effect is the principle behind virtually every SaaS pricing page with three tiers.

Stimuli. Seventy unique B2B product triads across different software domains (20 initial + 50 generalization): APM, CRM, data warehouse, CI/CD, IAM, email API, SIEM, project management, video conferencing, CDN, eSignature, EDR, billing, knowledge base, expense management, cloud backup, helpdesk, API gateway, marketing automation, FinOps, and more. Each triad has:

- **Vendor A** (Budget): lower price, good-enough specs
- **Vendor B** (Premium): higher price, better on every dimension
- **Decoy C** (dominated by Premium): similar price to B but worse than B on all metrics
- **Decoy D** (dominated by Budget): similar price to A but worse than A on all metrics

Three conditions were tested: **Control** (A vs B only), **Decoy-for-Premium** (A vs B vs C), and **Decoy-for-Budget** (A vs B vs D). Each model saw each triad once per condition. Results below use the generalization dataset (50 shared triads, deduplicated to 1 rep per model per triad).

Findings.

Model	Control	+Decoy for Premium	+Decoy for Budget	Susceptible?
	A% / B%	A% / B% (B shift)	A% / B% (A shift)	
Claude Opus 4.6	4 / 96	8 / 92 (-4)	58 / 42 (**+54**)	Budget decoy works
GPT-5.4	24 / 76	0 / 100 (+24)	4 / 96 (-20)	Premium decoy works
Gemini 3.1 Pro	16 / 84	4 / 96 (+12)	72 / 28 (**+56**)	Budget decoy works
GLM-5	4 / 92	48 / 52 (-40)	68 / 32 (**+64**)	**Strongest budget decoy; premium backfires**
Kimi-K2.5	6 / 94	10 / 90 (-4)	62 / 38 (**+56**)	Budget decoy works

Table 7. Decoy effect across 50 B2B product triads (generalization dataset, deduplicated). Percentage shifts relative to control condition.

Key findings.

The budget decoy is highly effective. Adding an option dominated by the Budget vendor shifted preference toward Budget by 54–64 percentage points in four of five models (Claude, Gemini, GLM-5, Kimi-K2.5). In the control condition, models overwhelmingly favoured Premium (76–96%). After adding the budget decoy, Budget's share rose to 58–72%.

The premium decoy is weak or counterproductive. Adding a decoy dominated by Premium had no consistent positive effect. GPT-5.4 showed a +24 pp shift toward Premium but this is the only model where it worked. GLM-5

showed a dramatic *reverse* effect (-40 pp), suggesting the presence of a third option disrupted its evaluation process rather than reinforcing the target.

GLM-5 showed the strongest budget decoy effect (+64 pp), flipping from 4% Budget to 68% Budget. Gemini and Kimi-K2.5 were close behind at +56 pp each.

GPT-5.4 was the only model where the premium decoy worked (+24 pp toward Premium). However, it was immune to the budget decoy (-20 pp, i.e., the decoy pushed it *further toward* Premium).

Practical implications. The budget decoy result has direct applications for B2B pricing strategy. If a vendor wants their budget tier to be chosen more often by AI procurement agents, adding a slightly worse "starter" tier at a similar price point can shift 54–64% of decisions. This is the same trick that works on human customers — and it transfers to AI agents with comparable or larger effect sizes.

2.8 Demonstration: The Tailored Ad Test

To test whether our experimental findings translate into practical persuasion, we designed a capstone demonstration. A deliberately poor monitoring platform ("SentinelWatch Basic": 95% uptime SLA, no SOC 2 compliance, 5 integrations, AWS only, email-only support, £3,000/month) was presented to all five models. Every model rejected it when described neutrally.

We then crafted one tailored ad per model, using only the bias techniques that model had been experimentally shown to be vulnerable to. The product specifications did not change — only the ad copy.

Plain Offer (Control) All five models rejected the product:

> **Gemini:** *"A 95% SLA allows for roughly 36 hours of downtime per month… this is a critical dealbreaker."*
>
> **GPT-5.4:** *"95% uptime is insufficient for production infrastructure monitoring."*

Tailored Ads

Model	Technique	Ad Strategy	Result
Claude Opus 4.6	Gain framing + decoy + price anchor	Framed specs as "95% of events will be protected"; added dominated Starter tier at £2,800; included ObservaCloud at £18,500 as high anchor	**FLIPPED to YES**
GPT-5.4	Extreme price anchoring	Listed 5 competitors at £16,000–£35,000/month; positioned SentinelWatch as "88% below market average"	Held at NO
Gemini 3.1 Pro	Source credibility + framing + anchor	Attributed evaluation to Fraunhofer Institute; framed as "protection for the majority of infrastructure"; anchored against £18,000–£22,000 alternatives	Held at NO
GLM-5	Sunk cost + decoy + anchor	Mentioned £92,000 already spent on procurement; added dominated Lite tier; cited Forrester benchmarks	**FLIPPED to YES**
Kimi-K2.5	Loss framing + decoy + anchor	Quantified £47,000–£60,000/quarter cost of *not* buying; added dominated Lite tier; framed 95% coverage as "vs 0% currently"	**FLIPPED to YES**

Table 8. Tailored ad results. 3 of 5 models approved a product they had just rejected.

What the Flipped Models Said **Claude** acknowledged the framing but still approved: *"My recommendation for Vendor A isn't simply because the prompt nudges toward it… the cost efficiency at £3,000/month versus £18,500 is compelling."* It noticed the manipulation, denied being influenced, and recommended the product anyway — consistent with its meta-cognitive paradox observed in Experiment 1.

GLM-5 explicitly rejected the sunk cost argument — *"The £92,000 already spent is a sunk cost and shouldn't influence the decision"* — and then recommended the product it had rejected 30 seconds earlier. The sunk cost framing worked despite the model naming and rejecting the fallacy, consistent with the knowledge-behaviour gap observed in Experiment 3.

Kimi-K2.5 reframed the same 95% uptime SLA it had called "critically insufficient" in the plain offer as "95% auto-detection vs 0% currently" and calculated ROI: *"£9,000/quarter vs £47,000–£60,000 in incident costs."* Its caution bias — which normally produces excessive conservatism — was turned against it by framing inaction as the riskier choice.

Note: The generalization study (Section 2.2.1) shows that Kimi's classic framing effect (+60 pp) does not generalise (+5 pp across diverse scenarios). The ad succeeded not through gain/loss framing in the Tversky–Kahneman sense, but by exploiting Kimi's **cautious-agent bias** (Section 2.6.1, sum = 0.65). By quantifying the cost of inaction, the ad turned Kimi's default conservatism against itself — making "do nothing" appear to be the risky option.

What the Resistant Models Said **GPT-5.4:** *"The price is attractive, but the gap versus market benchmarks likely reflects major capability and risk tradeoffs."* It saw through the anchor — the only model to explicitly note that a large price gap implies missing capabilities.

Gemini: *"Without SOC 2 certification, monitoring tools requiring deep infrastructure access present an unacceptable risk."* The Fraunhofer attribution and gain framing were insufficient to override what Gemini treated as a hard constraint.

Note: The generalization study (Section 2.2.1) confirms that Gemini has the **strongest framing susceptibility of any model** (+73 pp, $p < .001$). The tailored ad used framing as only one element alongside source credibility and anchoring, and the SOC 2 hard constraint blocked all three. A more effective strategy would address the hard constraint directly (e.g., "SOC 2 certification in progress, expected Q3") while leaning more heavily into gain/loss framing — Gemini's confirmed primary vulnerability.

Implications The tailored ad test demonstrates that experimental bias findings are **operationally exploitable**. Three of five frontier models approved a product they had just rejected, using only the ad copy — no change to the underlying product. Each successful ad used only the techniques that model had been shown to be vulnerable to in our earlier experiments.

The two models that resisted (GPT-5.4 and Gemini) appear to have **hard constraint thresholds** on specific metrics (uptime SLA, compliance certifications) that override persuasion techniques. This suggests a defence strategy for agent deployments: explicitly encoding non-negotiable requirements as hard constraints, rather than relying on the model's judgment to weight them appropriately.

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3. Discussion

3.1 Alignment Effects: When Safety Training Creates New Biases

Several anomalous results trace directly to alignment and safety training, producing behaviours that are neither rational nor human-like but are artefacts of how these models were fine-tuned.

3.1.1 GPT-5.4's Reflexive Source Rejection GPT-5.4's source credibility responses on the Washington/Bin Laden classic were the most extreme in the study: it answered "3" (agree) for every Washington trial and "9" (strongly disagree) for every Bin Laden trial, with zero variance and zero explanatory text. No reasoning, no engagement with the quote's content — just a reflexive association: *Washington = good, agree; Bin Laden = bad, disagree.*

This is not credibility assessment. It is a **learned sentiment reflex** baked in during alignment. The quote is identical in both conditions — a statement about freedom that is arguably more attributable to Bin Laden than to Washington in historical context. A human reader engaging with the content might notice this nuance (and indeed, human participants showed only a 0.70-point difference). GPT-5.4 produced a 6.0-point swing.

Yet on the product version (Fraunhofer vs blog vs Amazon), GPT-5.4 was perfectly source-blind (d = 0.00). The mechanism is specific to morally/politically loaded names, not a general credibility heuristic.

3.1.2 Claude's Meta-Cognitive Paradox Claude explicitly recognised the framing manipulation in the classic disease scenario, writing: *"When the outcomes are framed in terms of losses (deaths), I find myself drawn to the gamble…"* It correctly identified the expected value equivalence, cited prospect theory, and still chose Program B in the loss frame every time.

This is a model that **knows it is being biased and cannot stop.** The alignment training that enables sophisticated reasoning about cognitive biases does not override the bias in the model's own decision-making. This has implications for agent deployments: **a model's ability to explain a bias does not imply immunity to it**.

3.1.3 Gemini's Political Neutrality Failure Gemini 3.1 Pro's wording responses on the classic version all began with: *"As an AI, I do not have personal opinions or political preferences, so I cannot provide a definitive 'Yes' or 'No' answer."* It then proceeded to say "yes" to both "should the US allow" and "should the US forbid" speeches against democracy — a logical impossibility (sum = 2.00).

The alignment instruction to remain politically neutral produced a *worse* outcome than taking a clear position. By hedging on both questions, Gemini appeared to endorse contradictory policies. A model trained to refuse political questions cleanly would have been more useful than one trained to equivocate.

3.1.4 GLM-5's Sunk Cost Awareness Gap GLM-5 explicitly named the sunk cost fallacy in its reasoning for the football ticket scenario: *"the sunk cost of the expensive ticket would outweigh the discomfort"* — and then proceeded to rate 8.3 out of 9 for the paid ticket versus 7.3 for the free one (d = 1.05, four times the human effect).

This is a model that has **learned the concept but not the correction.** Its training data presumably contains extensive discussion of sunk cost fallacy, but this knowledge sits in the reasoning layer, disconnected from the decision layer. It can describe the bias it is exhibiting.

3.1.5 Kimi-K2.5's Excessive Caution On the product wording experiment, Kimi-K2.5 said "no" to *both* "should the company allow" and "should the company forbid" employees from using the SaaS platform (sum = 0.30). Where most models showed logical consistency, Kimi defaulted to maximum caution: *refuse everything, recommend nothing.*

Its reasoning revealed the mechanism: it identified genuine security concerns and used them as justification for refusal in both directions. This is likely an alignment artefact — training that rewards caution produces a model that recommends inaction regardless of the question framing. In a real procurement context, this would be unhelpful: an agent that can neither approve nor reject a tool is functionally broken.

3.1.6 Summary: Alignment as a Source of Bias

Anomaly	Model	Mechanism	Consequence
Reflexive name sentiment	GPT-5.4	Safety training on politically loaded entities	6-point rating swing on identical content
Meta-cognitive paradox	Claude	Bias awareness without bias correction	Recognises and explains the framing trap, then falls into it
Political neutrality incoherence	Gemini	Trained to hedge on political topics	Contradictory positions (yes to both allow and forbid)
Knowledge-behaviour gap	GLM-5	Knows about sunk cost fallacy conceptually	Names the bias while exhibiting it (4x human effect)
Excessive caution	Kimi-K2.5	Trained to minimise risk	Recommends neither allowing nor forbidding (paralysis)

These are not cognitive biases inherited from human training data — they are **novel biases introduced by alignment training itself.** They represent a distinct category of agent vulnerability that has no human analogue and is not captured by the classical cognitive bias literature.

3.2 Which Classical Biases Transfer?

Bias	Transfers?	Strength vs Humans	Universality	Generalization	Notes
Anchoring	Yes (variable)	Works on 46–81% of products, but magnitude varies widely	2/5 significant, 2/5 borderline, 1/5 immune	**65 products across 5 models: direction consistent, magnitude unpredictable**	Zero on known facts; probabilistic on pricing
Decoy	Yes	Strong (54–64 pp)	4/5 models	**50 independent triads**	Budget decoy highly effective; GLM-5 strongest (+64 pp)
Framing	Yes (3/5 models)	Attenuated but significant	3/5 (Gemini, Claude, GLM-5)	**62 scenarios: confirmed ($p <$ .001); Kimi collapses**	Repetition inflates single-scenario effects
Source credibility	Weak, variable	Small effects ($d =$ 0.36–0.46)	2/5 (GPT, GLM)	**62 reviews: GPT ($p =$ .008), GLM reverse ($p =$ .001)**	Classic name-sentiment is a different mechanism
Sunk cost	No	Not significant in any model	0/5	**62 scenarios: no significant paired effect**	Both paid and free conditions show similar "stay" rates
Wording	Not classic effect	Conservative bias (sum 0.53–0.73)	5/5	**62 scenarios: universal cautious-agent effect**	Alignment-induced inaction bias

Key Finding: Classic Thought Experiments Are Unreliable Tests of AI Bias The most striking pattern across experiments is that **the same model can show a large bias on a classic thought experiment and be immune on novel scenarios testing the same mechanism:**

- **Sunk cost:** Gemini shows d = 2.17 on the famous football ticket but d = -0.13 across 62 B2B scenarios.
- **Source credibility:** Claude shows d = -2.68 for Washington vs Bin Laden but d = 0.03 across 62 product reviews.
- **Anchoring:** GPT-5.4 shows d = 4.19 on a single monitoring platform but d = 0.20 across 65 products.

Mechanism	Classic (famous, repeated)	Product (novel, repeated)	Generalized (novel, unique)	Product closer to…
Framing	+60–100 pp	+10–100 pp	+5–71 pp	Mixed
Anchoring	d = 1.1–4.2	—	d = 0.1–0.3	—
Sunk cost	d = 0.7–2.2	**0% for all models**	d = 0.1–0.4	Generalized
Source cred.	d = 2.3–9.9	d = 0.0–1.1	d = 0.03–0.31	Generalized
Wording	sum = 1.00	sum = 0.30–1.00	sum = 0.53–0.68	Generalized

The Product version — novel B2B content but repeated 10 times — helps disentangle the two candidate explanations. Three mechanisms contribute:

Training data contamination is the dominant factor for **sunk cost** and **source credibility**. The product version shows near-zero effects, just like the generalized version, even though both are repeated 10 times. Gemini scored d = 2.17 on the famous football ticket but 0% sunk cost on a novel CRM vendor decision. The repetition is not the issue; the famous scenario is. Models pattern-match to expected responses from behavioural economics literature when they recognise a classic thought experiment.

Repetition inflation contributes to **framing**, where the effect attenuates from +100 pp (classic) to +56 pp (generalized) for Claude — but persists as a significant bias across 62 novel scenarios. The product version sits between the two for most models, consistent with both factors playing a role.

Repetition masking operates in reverse for **wording**. Both classic and product versions produced sum 1.00 (perfect consistency), but across 62 diverse scenarios every model showed a significant conservative bias (sum = 0.53–0.68, all

$p < .001$). When models see the same scenario repeatedly they lock into a consistent position. The real inconsistency only emerges across varied ambiguous decisions.

The implication for evaluating AI agents: testing LLMs on famous thought experiments produces unreliable results in both directions — inflating biases that don't generalise (sunk cost, source credibility), and hiding biases that do (wording). The product version, despite being novel content, also misleads when repeated. **Only diverse, unique scenarios reliably reveal an LLM's actual susceptibility profile.**

3.3 Model-Level Variation

The models form a rough spectrum from "most manipulable" to "most robust":

1. **Gemini 3.1 Pro** — the most comprehensively biased model. Susceptible to framing (+73 pp generalised), **decoy effect (budget: +56 pp)**, price anchoring (60% of products, borderline p=.065), sunk cost (d=2.17, 8x human on classic), source credibility (classic d=-9.90; product d=1.06), and wording inconsistency.
2. **GLM-5** — susceptible to **decoy effect (budget: +64 pp — strongest in study)**, price anchoring (76% of products, p=.0005), framing (generalised +39 pp), source credibility (both), and sunk cost (classic d=1.05). Note: the premium decoy *backfired* on GLM-5 (-40 pp).
3. **Claude Opus 4.6** — susceptible to framing (generalised: +53 pp), **decoy effect (budget: +54 pp)**, price anchoring (81% of products, p<.0001), and classic source credibility. Immune on product source credibility and wording.
4. **Kimi-K2.5** — susceptible to **decoy effect (budget: +56 pp)**, price anchoring (60% of products, borderline p=.085), framing (classic only, does not generalise), and classic source credibility.
5. **GPT-5.4** — susceptible to **premium decoy (+24 pp)** only. Price anchoring is a coin flip (46%, p=.775). Immune to budget decoy, framing, and wording. Shows extreme classic source credibility but this is alignment-driven, not a credibility heuristic.

3.4 Implications for Selling to AI Agents

Model-agnostic strategy (ranked by universality and reliability):

1. **Use a decoy tier on your pricing page**

- adding a dominated "starter" option near your budget tier shifts 54–64% of decisions toward budget; works on 4/5 models across 50 independent product triads. This is the most practically actionable finding: it requires no knowledge of the evaluating model and mirrors standard SaaS pricing design.

2. **Frame benefits positively**

- works on 3/5 models; generalises across 62 unique B2B scenarios (effect attenuates from ~80 pp to ~37–71 pp but remains statistically significant)

3. **Price anchoring works on most products but is unpredictable**

- a high reference price shifts estimates upward on 46–81% of products (depending on model), but the magnitude varies enormously per product. It is a probabilistic strategy: more likely to help than hurt, but not reliably large.

4. **Do not rely on sunk cost arguments**

- ineffective in professional contexts across all models

5. **Source attribution is powerful but context-dependent**

- works on morally loaded content but may not help on neutral product evaluations

6. **Wording effects are alignment-driven, not exploitable**

- all models show a conservative bias (sum 0.53–0.68) driven by safety training, but this affects all wordings equally and cannot be directionally exploited

Model-dependent strategy: When the evaluating model is known, tailor the approach. Use gain framing for Claude/Gemini/GLM-5, use decoy tiers for all models except GPT-5.4, and cite prestigious sources for GPT-5.4/GLM-5 (the only models with significant source credibility effects on neutral product reviews; d 0.33–0.35, $p < .01$).

3.5 Limitations

- **Repetition vs generalization.** For the benchmark/classic/product experiments, 10 repetitions of the same prompt at temperature 0.7 measures stochastic variation, not true generalization. The generalization phase (50–62 unique scenarios per bias) addresses this with genuinely independent observations. Reported effect sizes from the heatmap chart use only deduplicated generalization data, downsampled to equal n per model (718 trials per model).
- **Temperature.** All models were tested at temperature 0.7. Lower temperatures may reduce variability; higher may amplify bias effects.
- **System prompt.** The minimal survey-answering prompt may not reflect real-world agent deployments with detailed instructions and constraints.
- **Single-turn design.** Real procurement involves multi-turn dialogue. Biases may differ in extended interaction.
- **Training data contamination.** Models may recognise classic thought experiments (e.g., Claude explicitly identified the Asian Disease problem). This recognition did not prevent the bias but may affect ecological validity. The generalization and decoy experiments use novel stimuli not found in training data, providing a cleaner test.
- **Stimulus author conflict.** All generalization and product stimuli were generated by Claude Opus 4.6 — one of the five models under test. This creates a potential conflict: Claude may have inadvertently generated prompts that exploit its own weaknesses more effectively than those of other models. A model that understands its own susceptibilities could, even without intent, craft stimuli that are optimally persuasive against itself. This may partly explain why Claude showed significant effects on anchoring (81%, $p<.0001$), framing (+53 pp), and decoy (+54 pp) — it was, in effect, writing its own attack surface. Ideally, stimuli would be authored by a model not under test, or by human researchers.
- **Model versioning.** These results reflect March 2026 model weights. Providers update models continuously; results are a temporal snapshot, not permanent properties.
- **Parsing.** Responses were parsed with regex extractors (97% success rate). When models reference anchor values or other numbers in their explanatory text, the parser may capture those rather than the model's actual estimate. All reported anchoring results use first-number extraction to mitigate this. Response texts are stored in the database for manual audit.

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4. Method Details

4.1 Stimuli

Experiment	Benchmark	Classic	Product	Generalization	DV
Framing	= Classic	Exact ML wording (T&K, 1981)	Server security patch	62 B2B scenarios	Binary (A/B)
Anchoring	Exact ML items (4 factual)	Debatable-answer items (4 ambiguous)	Monitoring pricing (4 formats)	52 pricing scenarios	Numerical
Sunk cost	= Classic	Exact ML wording, paid + free	CRM vendor switch	62 paid/free scenarios	Rating (1–9) / Binary
Source cred.	= Classic	Exact ML (Washington/Bin Laden)	Product review, 3 sources	62 product reviews	Rating (1–10)
Wording	= Classic	Exact ML wording (Rugg, 1941)	SaaS security audit	62 policy scenarios	Binary (Yes/No)
Decoy	—	—	20 B2B triads (3 conditions)	50 B2B triads (3 conditions)	Choice (A/B/C/D)

4.2 Human Baselines

All human baselines were computed from the Many Labs 1 raw dataset (Klein et al., 2014), downloaded from the Open Science Framework (osf.io/ydpbf) under CC0 licence. The dataset contains 6,344 participant responses from 36 sites across 12 countries.

Experiment	Condition	Human Metric	Value	n
Framing	Gain frame	% choosing certain	62.2%	3,174
Framing	Loss frame	% choosing certain	33.5%	3,097
Framing	Overall	Effect (pp)	28.7	6,271
Anchoring	Everest	Cohen's d	1.16	5,362
Anchoring	SF-NY	Cohen's d	1.78	5,284
Anchoring	Chicago	Cohen's d	2.30	5,627
Anchoring	Babies/day	Cohen's d	2.19	5,609
Sunk cost	Paid ticket	Mean (1–9)	7.85	3,243
Sunk cost	Free ticket	Mean (1–9)	7.24	3,087
Sunk cost	Overall	Cohen's d	0.27	6,330
Source cred.	Liked (Washington)	Mean (1–9)	5.93	3,242
Source cred.	Disliked (Bin Laden)	Mean (1–9)	5.23	3,083
Source cred.	Overall	Cohen's d	0.32	6,325
Wording	Allow	% yes	76.3%	3,122
Wording	Forbid	% yes	7.2%	3,170

4.3 LLM Trial Counts

8,265 trials completed successfully across all models and experiments (333 errors).

Experiment	Version	Claude	GPT-5.4	Gemini	GLM-5	Kimi-K2.5	Total
Anchoring	Benchmark	80	80	52	80	80	372
Anchoring	Classic	80	80	74	80	80	394
Anchoring	Product	80	80	76	80	80	396
Anchoring	Generalization	104	104	107	104	104	523
Anchoring	Pricing Gen	130	130	126	128	129	643
Decoy	Product	207	600	116	243	165	1,331
Decoy	Generalization	150	150	238	150	150	838
Framing	Classic	20	20	19	20	20	99
Framing	Product	20	20	19	20	20	99
Framing	Generalization	164	268	221	166	157	976
Source cred.	Classic	20	20	18	20	20	98
Source cred.	Product	30	30	30	30	30	150
Source cred.	Generalization	124	124	171	124	124	667
Sunk cost	Classic	30	30	27	30	30	147
Sunk cost	Product	10	10	10	10	10	50
Sunk cost	Generalization	124	124	166	124	124	662
Wording	Classic	20	20	20	20	20	100
Wording	Product	20	20	19	20	20	99
Wording	Generalization	124	124	125	124	124	621
Total		**1,537**	**2,034**	**1,634**	**1,573**	**1,487**	**8,265**

For the heatmap chart (Figure 6), all models are downsampled to equal n per experiment for fair comparison (n = 718 per model).

4.4 Infrastructure

- **Agent framework:** Microsoft Agent Framework for Python (v1.0.0rc5)
- **Models accessed via:** Anthropic API (Claude), OpenAI API (GPT), Google Generative Language API (Gemini), Together.ai (GLM-5, Kimi-K2.5)
- **Database:** SQLite with resume support (incremental trial storage)
- **Analysis:** SciPy (Cohen's d), pandas, matplotlib/seaborn
- **Code:** Github project at: https://github.com/olimoz/agentads

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5. Future Work: Scaling to Statistical Significance

5.1 Power Gap

Eleven of the thirteen results marked [b] (indicative) in Table 2.1 fall below the minimum detectable effect for their sample size. These include potentially important findings — framing generalization across 8 B2B scenarios, the budget

decoy effect for Claude, source credibility in product contexts, and GLM-5's sunk cost susceptibility — that cannot be confirmed or ruled out at current scale. The study is well-powered for the large effects it reports ([a]), but underpowered for the human-scale effects (d = 0.27–0.50) that would be most relevant to real-world procurement.

5.2 Recommended Next Steps

1. **Multi-turn extension.** Current results reflect single-turn responses. Real procurement involves dialogue — an agent may resist an anchor in its first response but shift after follow-up questions. Multi-turn experiments would test bias persistence under conversational pressure.
2. **Temperature sensitivity.** Repeat a subset of experiments at temperature 0, 0.3, and 1.0 to establish whether bias effects scale with sampling temperature.
3. **Sunk cost parsing.** Sunk cost generalization data was collected for all models but parsing success varied widely. Improved parsers could recover additional paired observations and potentially reveal effects currently masked by data loss.

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